IoT Homework 2025/2026

Exercise 3: RFID

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1 Introduction

This report evaluates the collision resolution efficiency (η) of a Dynamic Frame ALOHA (DFA) RFID system with an initial population of $N = 4$ tags. The efficiency is defined as the ratio between the total number of initial tags and the expected total number of slots required for resolution (L_N^*):

$$\eta = \frac{N}{L_N^*} \tag{1}$$

We compute η across different initial frame sizes $r_1 \in \{1, 2, 3, 4, 5, 6\}$. We assume that from the second frame onward, the system dynamically sets the optimal frame size matching the remaining backlog ($r = N$). The baseline expected resolution times for smaller backlogs are given as $L_2 = 4$ and $L_3 = 51/8 = 6.375$ slots.

2 Preliminary Calculations: Baseline L_4

To evaluate an arbitrary r_1, we must first calculate the expected resolution time L_4 under optimal matching ($r = N = 4$). Let $P_r(S = i)$ be the probability of resolving i tags in a frame of size r. For $r = 4$, the total possible slot choices are $4^4 = 256$. The probabilities are derived combinatorially:

$$P_4(S = 4) = \frac{4!}{256} = \frac{3}{32} \tag{2}$$

$$P_4(S = 3) = 0 \quad \text{(Combinatorially impossible for } N = 4) \tag{3}$$

$$P_4(S = 2) = \frac{\binom{4}{2} \cdot \binom{4}{2} \cdot 2! \cdot \binom{2}{1}}{256} = \frac{144}{256} = \frac{9}{16} \tag{4}$$

$$P_4(S = 1) = \frac{\binom{4}{1} \cdot \binom{4}{1} \cdot \binom{3}{1}}{256} = \frac{48}{256} = \frac{3}{16} \tag{5}$$

$$P_4(S = 0) = 1 - \sum_{i=1}^{4} P_4(S = i) = \frac{5}{32} \tag{6}$$

Using the recursive relation for the expected duration:

$$L_4 = 4 + P_4(S = 0)L_4 + P_4(S = 1)L_3 + P_4(S = 2)L_2 \tag{7}$$

Substituting the known values ($L_2 = 4$, $L_3 = 51/8$):

$$L_4\left(1 - \frac{5}{32}\right) = 4 + \frac{3}{16}\left(\frac{51}{8}\right) + \frac{9}{16}(4) \implies L_4 = \frac{953}{108} \approx 8.824 \text{ slots} \tag{8}$$

3 Question 1: Efficiency Analysis for Variable r_1

For an arbitrary initial frame size r_1, the expected total time is:

$$L_4^*(r_1) = r_1 + P_{r_1}(S = 0)L_4 + P_{r_1}(S = 1)L_3 + P_{r_1}(S = 2)L_2 \tag{9}$$

We calculate $L_4^*(r_1)$ and $\eta(r_1)$ for $r_1 \in \{1, 2, 3, 4, 5, 6\}$:

- **Case $r_1 = 1$:** All tags collide in the single slot. $P_1(S = 0) = 1$.

$$L_4^*(1) = 1 + L_4 = 9.824 \implies \eta \approx \mathbf{0.407}$$

- **Case $r_1 = 2$:** Total configurations $2^4 = 16$. $P_2(S = 1) = \frac{4 \cdot 2 \cdot 1}{16} = 0.5$, $P_2(S = 0) = 0.5$.

$$L_4^*(2) = 2 + 0.5(8.824) + 0.5(6.375) = 9.599 \implies \eta \approx \mathbf{0.417}$$

- **Case $r_1 = 3$:** Total configurations $3^4 = 81$. $P_3(S = 2) = \frac{36}{81}$, $P_3(S = 1) = \frac{24}{81}$, $P_3(S = 0) = \frac{21}{81}$.

$$L_4^*(3) = 3 + \frac{21}{81}(8.824) + \frac{24}{81}(6.375) + \frac{36}{81}(4) = 8.954 \implies \eta \approx \mathbf{0.447}$$

- **Case $r_1 = 4$:** Matches the perfectly adapted baseline.

$$L_4^*(4) = L_4 = 8.824 \implies \eta \approx \mathbf{0.453}$$

- **Case $r_1 = 5$:** Total configurations $5^4 = 625$. $P_5(S = 2) = \frac{360}{625}$, $P_5(S = 1) = \frac{80}{625}$, $P_5(S = 0) = \frac{65}{625}$.

$$L_4^*(5) = 5 + \frac{65}{625}(8.824) + \frac{80}{625}(6.375) + \frac{360}{625}(4) = 9.038 \implies \eta \approx \mathbf{0.443}$$

- **Case $r_1 = 6$:** Total configurations $6^4 = 1296$. $P_6(S = 2) = \frac{720}{1296}$, $P_6(S = 1) = \frac{120}{1296}$, $P_6(S = 0) = \frac{96}{1296}$.

$$L_4^*(6) = 6 + \frac{96}{1296}(8.824) + \frac{120}{1296}(6.375) + \frac{720}{1296}(4) = 9.466 \implies \eta \approx \mathbf{0.423}$$

4 Question 2: Summary and Data Visualization

The numerical results computed across all scenarios are systematically arranged in Table 1. This architectural compilation highlights the structural variations in state probabilities and overall time overheads as a function of the first frame length. The empirical trajectory of the efficiency curve is visualized in Figure 1.

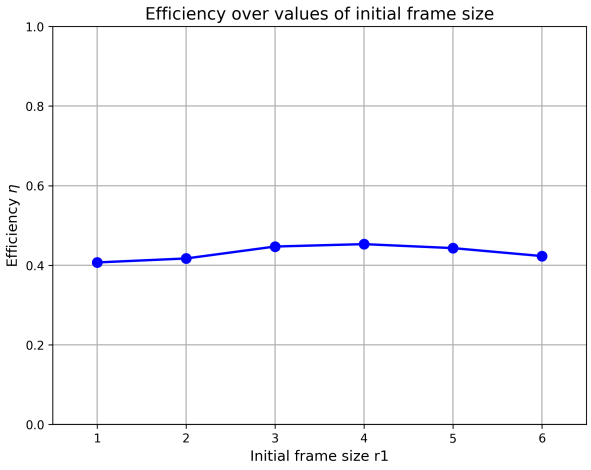


Figure 1: *System efficiency (η) as a function of the initial frame size (r_1).*

The curve clearly demonstrates a concave profile, rising steadily from $r_1 = 1$, reaching a distinct mathematical apex at $r_1 = 4$, and subsequently decaying as r_1 increases to 5 and 6.

Table 1: *DFA Arbitration Results for $N = 4$ Initial Tags*

r_1	$P_{r_1}(S=0)$	$P_{r_1}(S=1)$	$P_{r_1}(S=2)$	$P_{r_1}(S=4)$	L_4^* (slots)	Efficiency (η)
1	1.000	0.000	0.000	0.000	9.824	0.407
2	0.500	0.500	0.000	0.000	9.599	0.417
3	0.259	0.296	0.444	0.000	8.954	0.447
4	**0.156**	**0.188**	**0.563**	**0.094**	**8.824**	**0.453**
5	0.104	0.128	0.576	0.192	9.038	0.443
6	0.074	0.093	0.556	0.278	9.466	0.423

5 Question 3: Optimization and Performance Commentary

As proven by the numerical values in Table 1, the absolute maximum system efficiency is achieved exclusively at $r_1 = 4$, yielding $\eta \approx 0.4533$.

This behavior matches the fundamental throughput limits of Frame ALOHA protocols. Mathematically, the local slot efficiency inside a specific frame is maximized when the frame size r is perfectly identical to the number of competing un-resolved tags N ($r = N$).

- **Under-dimensioning ($r_1 < N$):** When $r_1 \in \{1, 2, 3\}$, the available slot capacity is insufficient for the population. This induces a high concentration of tags per slot, driving up the collision probability ($P_{r_1}(S = 0)$ scales up to 1.0 for $r_1 = 1$). Collisions destroy slot utility and force the system to waste entire initial frames, pushing heavy backlogs to subsequent cycles.

- **Over-dimensioning ($r_1 > N$):** When $r_1 \in \{5, 6\}$, the frame contains more slots than the actual tag population. While this lowers the collision rate and increases the chance of immediate resolution ($P_6(S = 4) \approx 0.2778$), it introduces an excessive number of un-utilized, empty slots. The time spent scanning these empty slots acts as an unnecessary delay overhead, lowering the aggregate efficiency.

Thus, setting $r_1 = 4$ provides the perfect operational balance, minimizing the combined penalty of empty slots and collision events.

6 Conclusion

This report explored the relationship between initial frame allocation and system performance in Dynamic Frame ALOHA systems. Through a rigorous combinatorial model, we proved that adjusting the initial frame length exactly to the tag population ($r_1 = N = 4$) yields the optimal resolution efficiency ($\eta \approx 45.33\%$). Deviations via under-provisioning lead to congestion, while over-provisioning introduces idle slot overhead. These findings validate the critical necessity of accurate backlog estimation algorithms in real-world RFID readers to dynamically maintain the $r = N$ matching condition across successive frames.